

Ali Ajvand

Data Scientist & BI Professional • Analytics Engineering, ML & End-to-End Data Products

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SUMMARY

I build the whole chain, not one link of it: raw source to ETL, model or semantic layer, then an API, dashboard, or interface the business actually opens. At JTI's Iran field-survey operation I built the reporting system from scratch and ran store-visit planning for the survey team, and I have built seven end-to-end data products (six solo, one with a teammate). I validate chronologically, audit for leakage before claiming a result, and benchmark against the incumbent process rather than another model: **9.36% WMAPE** forecasts against a 34.06% planner heuristic, and a credit policy that cut defaults **38%** at a higher approval rate. **2nd of 13** shortlisted teams at Hackathon Pol 4.

EXPERIENCE

Data Analyst, Project-Based • BehinBazar

Tehran • Apr 2025 – Nov 2025

Client: JTI (Japan Tobacco International), Iran retail-store field survey operations

Field survey data → Python + Excel → Power BI → daily reports on store coverage and team progress

- Built the reporting system for JTI's Iran field survey operation **from scratch** (no one had done this reporting before), delivering **daily reports** on store coverage and survey-team progress with Python, Excel, and Power BI.
- Managed store-visit assignment for the survey team (which stores each surveyor visited) and reported on each team member's progress.
- Built Folium geospatial tooling for store coverage and route planning, turning ad-hoc visit scheduling into a repeatable process.
- Automated parts of the daily reporting with Python and the Google Sheets API, replacing manual copy-paste steps.

END-TO-END DATA PRODUCTS

data layer → model → API → interface

Intelligent Medical Assistant • Clinical Decision-Support RAG Platform

[GitHub](#)

Document ingestion + OCR → Sentence-Transformers embeddings → FAISS retrieval → FastAPI service → Docker

- Built a retrieval-augmented decision-support platform as a deployable FastAPI service: content embedded with Sentence-Transformers (PyTorch) and served through semantic retrieval over a FAISS index.
- Included OCR so scanned documents flow through the same pipeline, and packaged the service with Docker for reproducible runs.
- Planned as a phased roadmap: symptom classifier → transformers on real data → OCR and semantic book search → drug data → doctor-patient requests → clinical-trial recommendations; the retrieval platform is the core deliverable.

Endoscopy Image Triage • Classifier with LLM-Explained Results

PyTorch • Flask • Llama 3.1 (Groq)

Image data → ResNet-50 transfer learning → Flask inference API → chat-style web interface → patient-facing explanation

- Took a 3-class classifier (Crohn's / ulcerative colitis / normal) all the way to a usable product: PyTorch model, Flask API, and a chat-style interface returning predicted class and confidence per uploaded image.
- Added an LLM layer (Llama 3.1 via Groq) that turns each prediction and its confidence into a short plain-language explanation, with rule-based fallback text when the LLM is unavailable so the interface never fails empty.
- Framed as a triage aid rather than a diagnosis: every result tells the patient to confirm with a specialist.

Credit Risk Decision Engine • Calibrated PD Scorecard & Approval Policy

public dataset, 30k clients • [GitHub](#)

Feature pipeline → calibrated PD model → cost-optimized policy → threshold table the business re-costs itself

- Cut the bad rate on the approved book from 15.9% to 9.8% (**38% fewer defaults**) while still approving 54.4% of applicants, by costing a 5:1 false-approval to false-decline policy instead of a naive PD < 0.50 cutoff.
- Calibrated PD on 41 behavioral features: ROC-AUC 0.780, PR-AUC 0.554, KS 0.427, Brier 0.135, against 0.703 AUC for the single-rule baseline; the riskiest decile defaults at 70.0% versus 22.1% portfolio-wide (3.17× lift).
- Shipped the decision layer, not just the model: a threshold table the business re-costs when loss assumptions change; sex, education, marital status, and age are excluded from features and kept for adverse-impact monitoring.

Hackathon Pol 4 • Hospitality Demand Forecasting • 2nd of 13 shortlisted teams

Team of two • 2026

AI Innovation Factory × Jobama • 72-hour competition

Real booking data → nowcasting and seasonality features → 4-model ensemble → 2-4 week early-warning signal

- With one teammate, built a time-series ensemble (LightGBM, XGBoost, CatBoost, HistGradientBoosting) reaching **82% accuracy** on real hotel-booking data across 321 Iranian cities, from scratch in 72 hours.
- Combined accrual-based nowcasting with calendar seasonality and geographic features, framed as an early-warning signal: when to move inventory and pricing 2 to 4 weeks ahead of a demand peak.

FORECASTING, ANALYTICS & BI

Demand & Sales Forecasting · Hierarchical Retail Time Series

[GitHub](#)

[Hierarchical sales data](#) → [walk-forward validation](#) → [per-store forecasts](#) → [replenishment thresholds](#) operations can act on

- Cut forecast error **72%** versus the planner heuristic (34.06% → 9.36% WMAPE), moving **98.3% of stores** inside the 15% threshold where replenishment can run without human review.
- Forecasted 844,338 open store-days across three non-overlapping 42-day walk-forward folds, training data always ending before the fold begins; held bias at -1.9% against a ±5% target.
- Published per-store error (median 8.7%, p90 11.9%) so safety stock is set store by store instead of one padded number chain-wide.

EV & Automotive Market Intelligence · Power BI Star Schema & Price Regression

public data · [GitHub](#)

[Raw registrations and listings](#) → [star schema](#) → [DAX measures](#) → [drill-through Power BI report + pricing model](#)

- Modeled 289,555 EV registrations in a Power BI star schema with drill-through and a documented measure layer: 80.7% BEV share and 69.3% fleet concentration in the top three counties, pointing charging-infrastructure spend at where the fleet actually is.
- Priced 15,409 used-car listings on a nameplate-grouped split, where an explainable linear regression (R^2 0.83, 19.3% MAPE) beat gradient boosting (20.8%) and more than halved the 41.8% error of the brand-median rule pricing teams apply by hand.
- Caught a leakage failure before it shipped: an apparent ROC-AUC of 1.00 collapsed to 0.66 once drivetrain-derived proxy features were removed, killing a claim that would not have survived production.

Delivery Performance Analytics · Predictive Logistics & Decision Policies

public Olist dataset · [GitHub](#)

[As-of joins keyed to purchase time](#) → [five models](#) → [calibrated operating policies](#) → [cost-sensitivity tables](#)

- Predicted late delivery, freight cost, and promise dates across ~99,000 e-commerce orders (3,000 sellers, 27 states) using purchase-time information only, so every model is deployable at checkout rather than after the fact.
- Engineered seller, route, and customer-history features through as-of joins, then proved the pipeline clean with a 39-feature leakage audit and 34 automated tests.
- Turned five models into policies operations can run, including capacity-constrained expedite queues and quantile promise dates, each with isotonic calibration, cost-sensitivity tables, and an incumbent baseline to beat.

Also built: Portfolio site (React, Vite, deployed with GitHub Actions) · Power BI e-commerce sales dashboard on MySQL · LangChain + FAISS RAG pipeline over structured vehicle data (Hugging Face embeddings) · K-means customer segmentation · FP-Growth market-basket analysis · A/B testing and ad-spend analysis. [Full case studies](#)

TECHNICAL SKILLS

Data & Modeling: SQL, PostgreSQL, MySQL, SQL Server, T-SQL, DuckDB, Parquet, ETL/ELT, dimensional modeling (star schema), data quality checks

BI & Visualization: Power BI (star schema, DAX, drill-through, KPI dictionaries), Plotly, Seaborn, Matplotlib, Folium. Familiar: Tableau, Metabase, Streamlit

Python & Machine Learning: Python, pandas, scikit-learn, XGBoost, LightGBM, CatBoost, forecasting, classification, regression, clustering, feature engineering, SHAP, probability calibration (isotonic, Platt)

Deep Learning & LLM Apps: PyTorch, transfer learning, Hugging Face Transformers, Sentence-Transformers, FAISS, LangChain, RAG, LLM APIs (Groq)

Validation: walk-forward and grouped cross-validation, leakage auditing, A/B testing, hypothesis testing, cost-sensitive evaluation

Shipping: FastAPI, Flask, Docker, Git, GitHub Actions CI/CD, pytest, ruff, mypy, reproducible pipelines

EDUCATION, TRAINING & LANGUAGES

B.Sc. Computer Engineering · Islamic Azad University, Science & Research Branch

2026 · GPA 16/20

M.Sc. entrance examination (Computer Engineering) completed in 2026; program placement pending.

Training: Data Scientist Bootcamp, Daneshkar (100+ hrs, 2025) · LangChain for LLM Application Development, freeCodeCamp (2026) · Business Intelligence Bootcamp, Udemy (2023) · Data Analyst Bootcamp, Alex The Analyst (2023)

Languages: Persian (native) · English (professional reading and writing)